이름 없는 노트북

1. $\pi_{cid, aid, pid}$(Customers ⋈ Agents ⋈ product)

2. $\pi_{cid, aid, pid}$(Customers.city = Agents.city ∨ Agents.city = products.city) ∨ (customers.city ≠ Agents.city ≠ Agents.city ≠ products.city).

3. π_{pname}($\sigma_{Customers.city=NZ \wedge Agents.city=SH}$(customers × Agents × products) ⋈ $\pi_{cid,aid,pid}$(orders))

4. π_{aname}($\pi_{aid,pid}$(orders) − π_{pid}($\sigma_{cid=c002}$(orders)) ⋈ Agents

5. $\pi_{cid,aid,pid}$($\sigma_{Customers.city = Agents.city \vee Agents.city}$ ∨ Agents.city = products.city ∨ customers.city = products.city.

6. π_{aid}($\sigma_{Customer.city=SH \wedge Orders.Qty \times products.price > 50}$ (Orders ⋈ products ⋈ Customers.

7. π_{cid}(Orders) − π_{cid}($\sigma_{Orders.cid = Orders2.cid \wedge}$

$\cup$ customers.city = Agents.city $\cup$ Agents

$\cup$ Agents.city = products.city $\cup$ customers.city

6. $\pi_{aid}(\sigma_{customer.city = SF \wedge Orders.Qty \times products.price > 50}$
(Orders $\bowtie$ products $\bowtie$ Customers.

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7. $\pi_{cid}(Orders) - \pi_{cid}(\sigma_{Orders.cid = Orders2.cid \wedge}$
$Orders.aid \neq Orders2.aid(Orders \times Orders))$

90%

6.

1) $\pi_{sno}(\sigma_{jno='j1'}(spj))$

2) $\pi_{sno}(\sigma_{jno='j1'}(spj) \cap \sigma_{pno='p1'}(spj))$

3) $\pi_{sno}(\sigma_{jno='j1'}(spj) \bowtie \sigma_{color='红'}(p))$

4) $\pi_{jno}(j) - \pi_{jno}(\pi_{sno}(\sigma_{city='天津'}(s)) \bowtie \pi_{pno}(\sigma_{color='红'}(p)) \bowtie \pi_{sno,pno,jno}(spj))$

5) $\pi_{jno,pno}(spj) \div \pi_{pno}(\sigma_{sno='s1'}(spj))$

7.试述等值连接与自然连接的区别和联系。

自然连接一定是等值连接，但等值连接不一定是自然连接。

等值连接要求相等的分量，不一定是公共属性，但自然连接要求相等的分量必须是公共属性。

8.关系代数的基本运算有哪些？如何用这些基本运算来表示其他运算？

在关系代数的八种运算中并，差，笛卡尔积，投影和选择为基本运算。